

TAK Server - AD/LDAP

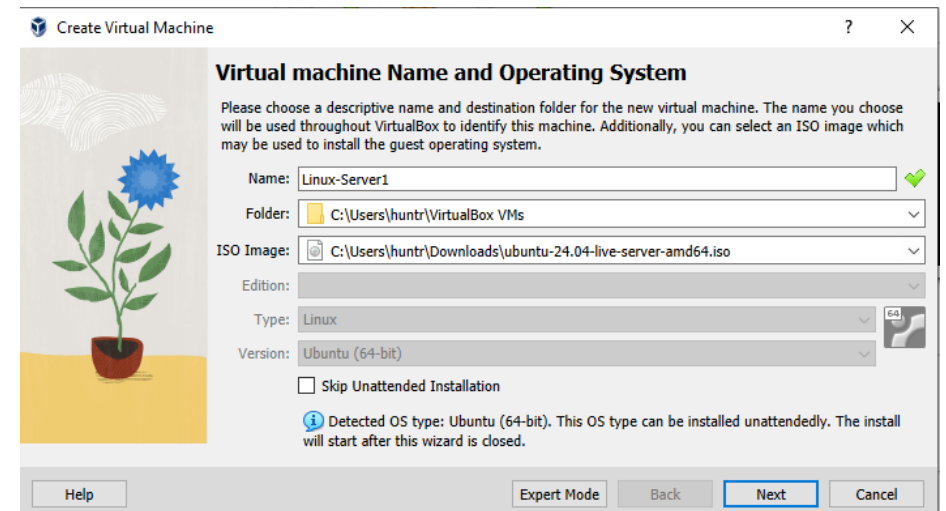
You should have a TAK Server fully stood up to proceed.

TAK Server - AD/LDAP Integration

- Generally, with either AD or LDAP domains you have an established certificate authority and then an authentication method, either a CAC (common access card)/token or username and password.
- (Required) - Create a TAK service account to allow the server to query information from the AD/LDAP domain.
- (Recommended) - Create a certificate signing request and recreating the java and pkcs12 trust stores to have our server signed by the domain issuing authority.
- (Optional) - Configure certificate autoenrollment using AD (Active Directory) Certificate Services.

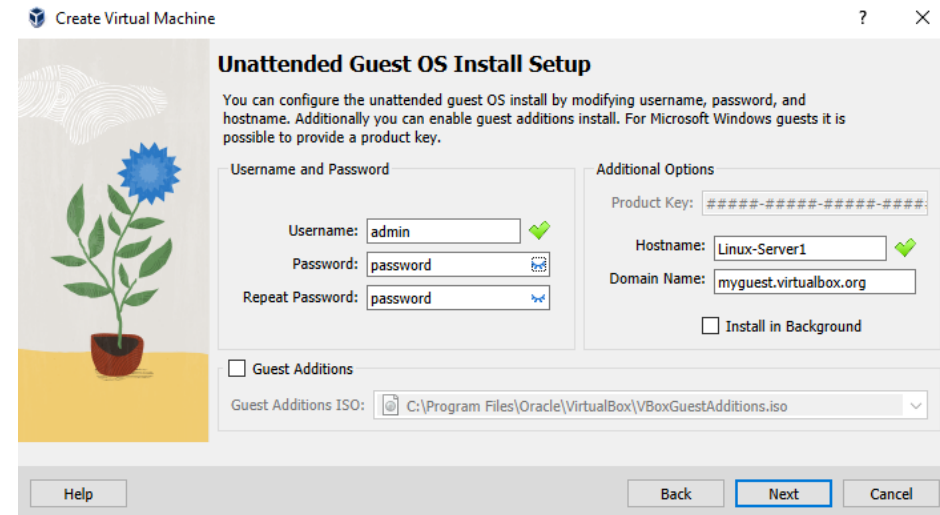
TAK Server - AD/LDAP Integration - LDAP Server Setup

- Create a Ubuntu (or your choice of OS).
 - Download your choice of OS boot iso.
 - <https://ubuntu.com/download/desktop>
 - Within Oracle Virtual Machine select “New”
 - Fill in the appropriate fields:
 - Name: <name your server>
 - Folder: <change location if needed>
 - ISO Image: <select the downloaded image>
 - Select “Next”



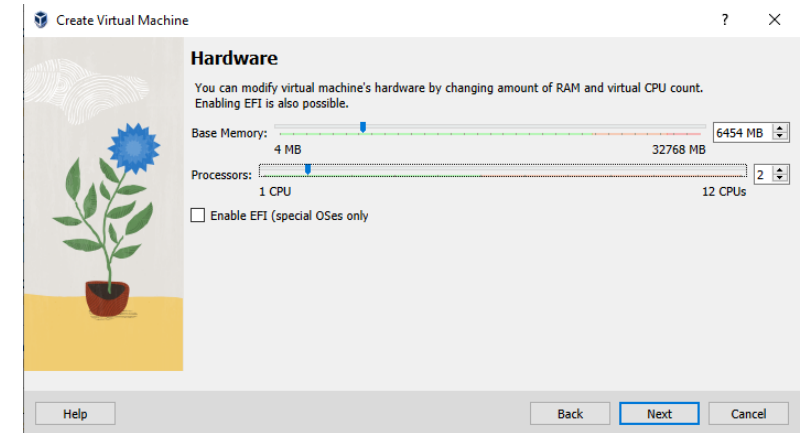
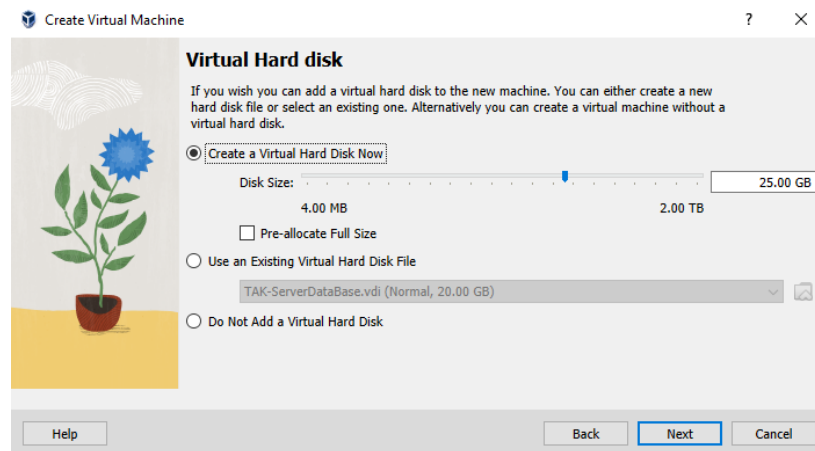
TAK Server - AD/LDAP Integration - LDAP Server Setup

- Here you can change the username and password to whatever you'd like:
 - Username: admin
 - Password: password
- You can also change the name of the host, and the domain name is you desire.
- When done select “Next”

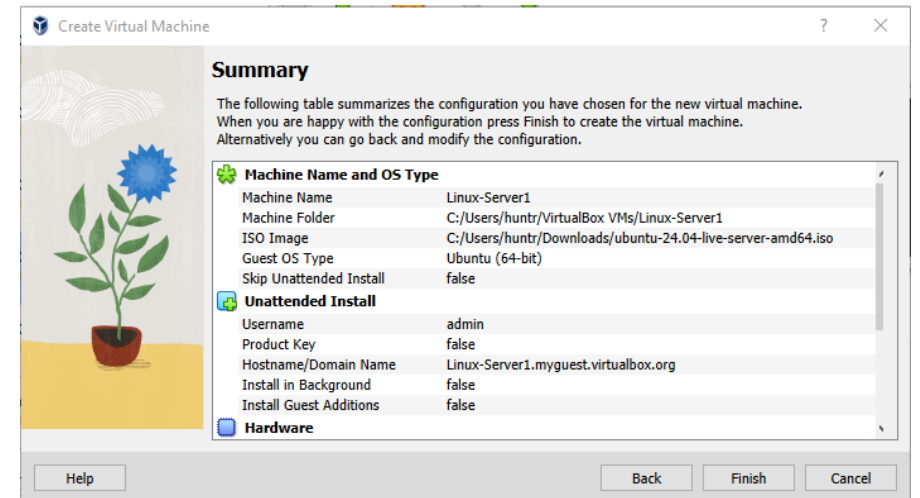


TAK Server - AD/LDAP Integration - LDAP Server Setup

- Allocate the desired memory and CPU cores:
- Ensure there is enough memory allocated to the disk:



- Select “Finish”



TAK Server - AD/LDAP Integration - User Setup

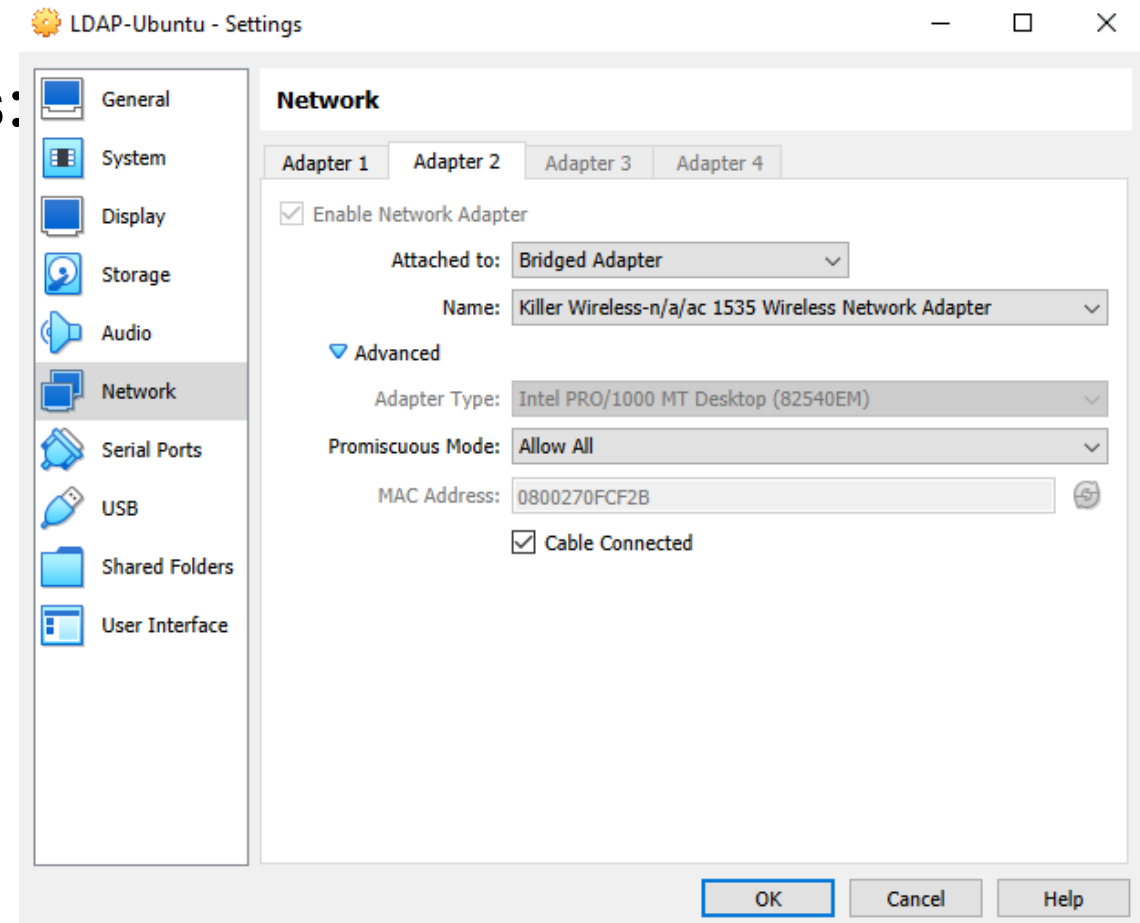
- Walk through the setup process for the OS.
- Once loaded, login.
 - If using Ubuntu provided above following these instructions:
 - Open a terminal and try using sudo: `sudo apt-get update`
 - If you get an error saying “the user is not in the sudoers file” do the following:
 - Enter root user:
 - `su -`
 - enter your password
 - promote the user:
 - `sudo adduser <username> sudo`
 - switch back to the user:
 - `su <username>`

```
tak@TAKDBServer:~$ sudo apt-get update
[sudo] password for tak:
tak is not in the sudoers file. This incident will be reported.
tak@TAKDBServer:~$ su -
Password:
root@TAKDBServer:~# sudo adduser tak sudo
Adding user `tak' to group `sudo' ...
Adding user tak to group sudo
Done.
root@TAKDBServer:~# su tak
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

tak@TAKDBServer:/root$ cd ~
tak@TAKDBServer:~$ sudo apt-get update
[sudo] password for tak:
Hit:1 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
tak@TAKDBServer:~$ S
```

TAK Server - AD/LDAP Integration -Network Setup

- Power off the server.
- In the VM go to the server's settings:
 - Display - > Remote Display:
 - Enable “Enable Server”
 - Enable “Allow Multiple Connections”
 - Network -> Adapter 2:
 - Enable “Enable Network Adapter”
 - Attached to “Bridged Adapter”
 - Advanced:
 - Promiscuous Mode: Allow All
 - Select “ok” and start the server



TAK Server - AD/LDAP Integration - Network Setup

- Login and open terminal:
 - Install OpenSSH:
 - `sudo apt install openssh-server -y`
 - Install network tools:
 - `sudo apt install net-tools`
 - You should now be able to ssh into the server from your host machine.
 - `ssh <username>@<IP address>`
 - Find the IP address with: `ifconfig`

```
C:\Users\huntr>ssh tak@192.1.168.115
C:\Users\huntr>ssh tak@192.168.1.115
The authenticity of host '192.168.1.115 (192.168.1.115)' can't be established.
ECDSA key fingerprint is SHA256:xnFEEjDjnxM37C3HCYWLESihLOGFmjnEnaj4clQIzkg.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.115' (ECDSA) to the list of known hosts.
tak@192.168.1.115's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 6.5.0-45-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

562 updates can be applied immediately.
373 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

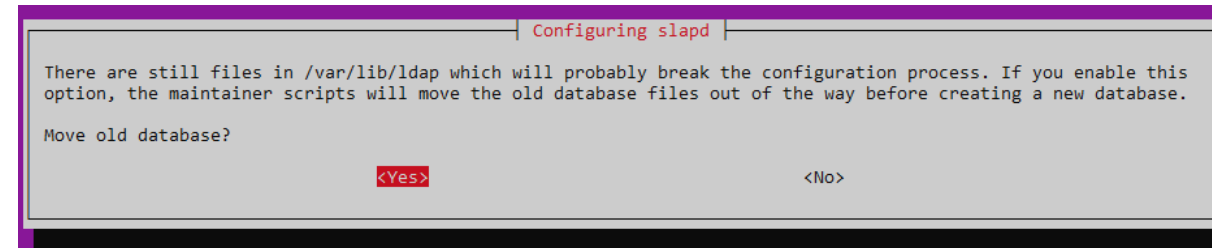
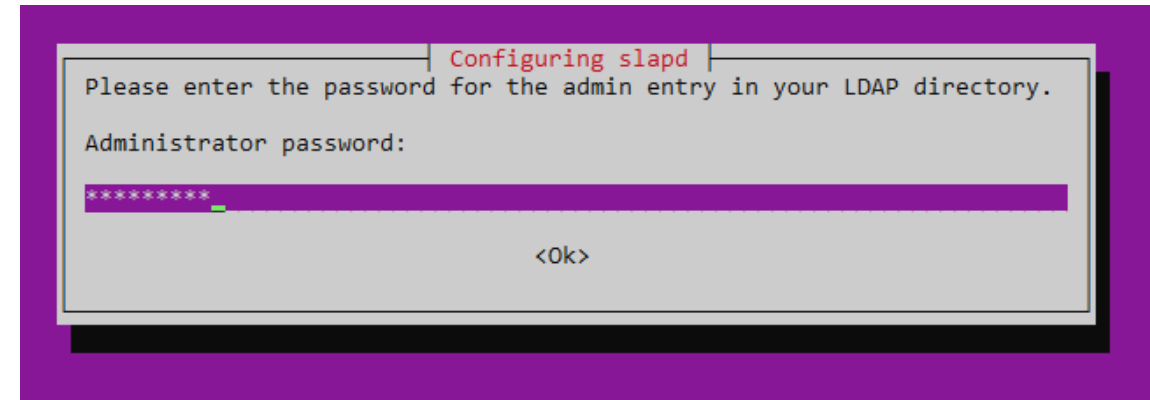
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

tak@TAKDBServer:~$
```

TAK Server - AD/LDAP Integration - Setup

- Update your system:
 - `sudo apt update`
- Install the OpenLDAP server and utilities:
 - `sudo apt install slapd ldap-utils -y`
 - You will be prompted to enter a password:
 - Password1
- Configure OpenLDAP
 - `sudo dpkg-reconfigure slapd`
 - Omit OpenLDAP server Configuration: No
 - DNS Domain Name: tak.gov
 - Organization Name: Test.tak.gov
 - Password: Password1
 - Re-Enter Password: Password1
 - Remove Database when slapd is purged: NO
 - Move old data: Yes



TAK Server - LDAP Integration - Setup

- Check the status:
 - `sudo systemctl status slapd`
- If the service is not running, start it:
 - `sudo systemctl start slapd`
- To restart the service:
 - `sudo systemctl restart slapd`
- To see the current backend configuration:
 - `sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config dn`

```
takadmin@LDAP-Ubuntu2:~$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config dn
dn: cn=config
dn: cn=module{0},cn=config
dn: cn=schema,cn=config
dn: cn={0}core,cn=schema,cn=config
dn: cn={1}cosine,cn=schema,cn=config
dn: cn={2}nis,cn=schema,cn=config
dn: cn={3}inetorgperson,cn=schema,cn=config
dn: olcDatabase={-1}frontend,cn=config
dn: olcDatabase={0}config,cn=config
dn: olcDatabase={1}mdb,cn=config
takadmin@LDAP-Ubuntu2:~$
```

TAK Server - LDAP Integration - Setup

- Search for the following schemas, if have them you don't need add them again:
 - **sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=schema,cn=config dn**
- Import Additional Schemas:
 - **core.ldif** defines the fundamental object classes and attributes that are required by LDAP directories.
 - **sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/ldap/schema/core.ldif**
 - **cosine.ldif** includes additional object classes and attributes used for compatibility with x.500 directories.
 - **sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/ldap/schema/cosine.ldif**
 - **nis.ldif** extends the core schema with attributes and object classes used in Unix environments for presenting networked information.
 - **sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/ldap/schema/nis.ldif**
 - **inetorgperson.ldif** extends the organizationalPerson object class with attributes that are commonly used for internet and organizational applications.
 - **sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/ldap/schema/inetorgperson.ldif**

TAK Server - LDAP Integration - Setup

- Add base schema and entries:
 - Create a LDIF file named base.ldif with the following content:
 - To create a file: **vim base.ldif**
 - Add the entries to the LDAP directory (replace tak.gov with your domain):
 - **sudo ldapadd -x -D cn=admin,dc=tak,dc=gov -W -f base.ldif**
 - Verify the entries:
 - **sudo ldapsearch -x -LLL -b dc=tak,dc=gov 'ou=People' dn**
 - **sudo ldapsearch -x -LLL -b dc=tak,dc=gov 'ou=Groups' dn**

```
dn: ou=People,dc=tak,dc=gov
objectClass: organizationalUnit
ou: People

dn: ou=Groups,dc=tak,dc=gov
objectClass: organizationalUnit
ou: Groups
```

```
takadmin@LDAP-Ubuntu2:~$ sudo ldapsearch -x -LLL -b dc=tak,dc=gov 'ou=People' dn
dn: ou=People,dc=tak,dc=gov

takadmin@LDAP-Ubuntu2:~$ sudo ldapsearch -x -LLL -b dc=tak,dc=gov 'ou=Groups' dn
dn: ou=Groups,dc=tak,dc=gov

takadmin@LDAP-Ubuntu2:~$
```

TAK Server - LDAP Integration - Setup

- Change LDAP Administrator password:
 - Search the LDAP configuration database to see how the DN is configured:
 - **sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config "(olcRootDN=*)" olcRootDN**
 - Generate a new hashed password:
 - **sudo slappasswd**
 - Enter a password and when complete copy the generated SSHA password.
 - Create a new LDIF file named "change_password.ldif" with the following content:
 - Change the SSHA password to the one you generated.
 - Apply the LDIF file to change the password:
 - **sudo ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f change_password.ldif**

```
takadmin@LDAP-Ubuntu2:~$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config "(olcRootDN=*)" olcRootDN
dn: olcDatabase={0}config,cn=config
olcRootDN: cn=admin,cn=config

dn: olcDatabase={1}mdb,cn=config
olcRootDN: cn=admin,dc=tak,dc=gov
```

```
dn: olcDatabase={1}mdb,cn=config
changetype: modify
replace: olcRootPW
olcRootPW: {SSHA}D7MVZecS7ZKsh1IwboR80Hld6ToxF4oX
~
```

TAK Server - LDAP Integration - Setup

- Adding organizational units (OU)
 - Create a new LDIF file called add_org_unit.ldif
 - Add the following to it:

```
dn: ou=Engineering,dc=tak,dc=gov
objectClass: organizationalUnit
ou: Engineering
```
 - Add the OU:
 - **ldapadd -x -D "cn=admin,dc=tak,dc=gov" -W -f add_org_unit.ldif**
 - Verify the OU has been added:
 - **ldapsearch -x -LLL -b dc=tak,dc=gov "(ou=Engineering)" dn**

```
takadmin@LDAP-Ubuntu2:~$ ldapsearch -x -LLL -b dc=tak,dc=gov '(ou=Engineering)' dn
dn: ou=Engineering,dc=tak,dc=gov
```

TAK Server - LDAP Integration - Setup

- Adding users:
 - To add a new user to the Engineering OU that we just created, create a new LDIF file and add the following:
 - You can create a new SSHA password to use.
 - You can also change the name and other attributes to whatever you'd like.
 - Once done save and exit the file then add the user to the Engineering OU:

```
dn: uid=jdoe,ou=Engineering,dc=tak,dc=gov
objectClass: inetOrgPerson
objectClass: posixAccount
cn: John Doe
sn: Doe
givenName: John
displayName: John Doe
uid: jdoe
uidNumber: 1001
gidNumber: 5000
homeDirectory: /home/jdoe
loginShell: /bin/bash
userPassword: {SSHA}D7MVZecS7ZKsh1IwboR80Hld6ToxF4oX
mail: jdoe@example.com_
```

- **ldapadd -x -D "cn=admin,dc=tak,dc=gov" -W -f add_user_to_engineering.ldif**

```
takadmin@LDAP-Ubuntu2:~$ ldapadd -x -D "cn=admin,dc=tak,dc=gov" -W -f add_user_to_engineering.ldif
Enter LDAP Password:
adding new entry "uid=jdoe,ou=Engineering,dc=tak,dc=gov"

takadmin@LDAP-Ubuntu2:~$
```

- Verify the user was added:
 - **ldapsearch -x -LLL -b dc=tak,dc=gov "(uid=jdoe)" dn**

TAK Server - LDAP Integration - Setup

- Modifying Entries:

- Create a LDIF file called modify_user.ldif with the following:
 - This changes the users email address.
- User the modify command to modify the user:
 - **ldapmodify -x -D "cn=admin,dc=tak,dc=gov" -W -f modify_user.ldif**
- Verify the changes:
 - **ldapsearch -x -LLL -b dc=tak,dc=gov "(uid=jdoe)" mail**

```
dn: uid=jdoe,ou=Engineering,dc=tak,dc=gov
changetype: modify
replace: mail
mail: john.doe@example.com
```

```
takadmin@LDAP-Ubuntu2:~$ ldapsearch -x -LLL -b "dc=tak,dc=gov" "(uid=jdoe)" mail
dn: uid=jdoe,ou=Engineering,dc=tak,dc=gov
mail: john.doe@example.com
```

TAK Server - LDAP Integration - Setup

- Searching entries:
 - Search for all users:
 - **ldapsearch -x -LLL -b "dc=tak,dc=gov" "(objectClass=inetOrgPerson)"**
 - Searching for a specific user:
 - **ldapsearch -x -LLL -b "dc=tak,dc=gov" "uid=jdoe"**

```
takadmin@LDAP-Ubuntu2:~$ ldapsearch -x -LLL -b "dc=tak,dc=gov" "(objectClass=inetOrgPerson)"
dn: uid=jdoe,ou=Engineering,dc=tak,dc=gov
objectClass: inetOrgPerson
objectClass: posixAccount
cn: John Doe
sn: Doe
givenName: John
displayName: John Doe
uid: jdoe
uidNumber: 1001
gidNumber: 5000
homeDirectory: /home/jdoe
loginShell: /bin/bash
mail: john.doe@example.com
```

TAK Server - LDAP Integration - Setup

- Changing Passwords:

- Create a new password:

- **sudo slappasswd**

- Create a new LDIF file called change_user_password.ldif with the following content:

```
dn: uid=jdoe,ou=Engineering,dc=tak,dc=gov
changetype: modify
replace: userPassword
userPassword: {SSHA}Jorj79MirFbUAhaMV1FC2SpcScXw+elq
~
```

- Commit the changes:

- **ldapmodify -x -D "cn=admin,dc=tak,dc=gov" -W -f change_user_password.ldif**

```
takadmin@LDAP-Ubuntu2:~$ ldapmodify -x -D "cn=admin,dc=tak,dc=gov" -W -f change_user_password.ldif
Enter LDAP Password:
modifying entry "uid=jdoe,ou=Engineering,dc=tak,dc=gov"
```

TAK Server - LDAP Integration - Setup

- Backup and restore:
 - To backup the LDAP database:
 - **sudo slapcat -n 1 -l backup.ldif**
 - To restore the LDAP database:
 - Stop the service:
 - **sudo systemctl stop slapd**
 - Restore the database:
 - **sudo slapadd -n 1 -l backup.ldif**
 - To start the LDAP service:
 - **sudo systemctl start slapd**
 - To verify the restoration:
 - **ldapsearch -x -LLL -b "dc=tak,dc=gov" "(objectClass=*)"**

TAK Server - LDAP Integration - Setup

- Access Control Lists (ACLs) allow for fine-grained control over who can read, write and manage directory entries and attributes.
- ACLs are critical for securing your LDAP directory and ensuring that only authorized users can access or modify sensitive information.
- ACLs are defined in the OpenLDAP configuration files and are structured as follows:

```
olcAccess: to <target>  
    by <access control rule>  
    by <access control rule>  
    ...
```

- Target: specifies the directory entries or attributes the ACL applies to.
- Access Control Rule: Defines who can perform what actions on the target.

TAK Server - LDAP Integration - Setup

- Example of ACL:

```
olcAccess: to dn.subtree="ou=users,dc=example,dc=com"  
  by self write  
  by dn.base="cn=admin,dc=example,dc=com" write  
  by * read
```

- Add the ACL file:

```
ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f acl.ldif
```

TAK Server - LDAP Integration - Setup

- Deleting Entries:

- Command to delete a user:

```
ldapdelete -x -D "cn=admin,dc=example,dc=com" -W "uid=jdoe,ou=Engineering,dc=example,dc=com"
```

- Verify the user has been deleted:

```
ldapsearch -x -LLL -b "dc=example,dc=com" "(uid=jdoe)" dn
```

- Command to delete an OU:

```
ldapdelete -x -D "cn=admin,dc=example,dc=com" -W "ou=Engineering,dc=example,dc=com"
```

- Verify the OU has been deleted:

```
ldapsearch -x -LLL -b "dc=example,dc=com" "(ou=Engineering)" dn
```

TAK Server - LDAP Integration - Setup

- Install Apache Directory Studio:
 - <https://directory.apache.org/studio/>
- In the top left select: “File” -> “New” -> LDAP Browser -> LDAP Connection.
 - Give it a name and then enter in the IP address of the server.
 - Select “Next”
 - Authentication:
 - Bind DN: cn=admin,dc=tak,dc=gov
 - Bind Password: password
 - Select “Check Authentication”
 - If successful select “Finish”

TAK Server - LDAP Integration - Setup

- Open up you TAK Server CoreConfig.xml file from the tak user:
 - vim /opt/takCoreConfig.xml
- Make the following changes:

```
</network>
<auth default="ldap" x509groups="true" x509addAnonymous="false">
  <ldap url="ldap://192.168.1.117:389" userstring="uid={username},ou=People,dc=tak,dc=gov" updateinterval="60" groupNameExtractorRegex="CN=(.*)" style="DS"
serviceAccountDN="cn=admin,dc=tak,dc=gov" serviceAccountCredential="password" groupObjectClass="groupOfNames" userObjectClass="inetOrgPerson" groupBaseRDN="dc=tak,dc=gov"
userBaseRDN="ou=People,dc=tak,dc=gov"/>
</auth>
<submission ignoreStaleMessages="false" validateXml="false"/>
<subscription reloadPersistent="false"/>
<repository enable="true" numDbConnections="200" primaryKeyBatchSize="500" insertionBatchSize="500">
```

- Validate the changes in the tak use:
 - cd ~
 - ./validateConfig.sh
- Restart the server (from outside the tak user):
 - sudo systemctl restart takserver

TAK Server - LDAP Integration - Connecting our TAK Server

- From our server dashboard the configuration menu should look as follows:
- In our TAK Server dashboard: Hamburger menu -> Configuration -> Security and Authentication
 - Select “Edit Authentication”
 - URL: <ldap://192.168.1.115:389>
 - User String: cn={username},ou=People,dc=tak,dc=gov
 - Update Intervals: 60
 - Service Account DN: cn=admin,dc=tak,dc=gov
 - Security Account Password: Password1
 - Group Base RDN: dc=tak,dc=gov

Security Configuration

Keystore File: certs/files/takserver.jks
Truststore File: certs/files/truststore-intermediate-ca.jks
TLS Version: TLSv1.2
x509 Groups: true
x509 Add Anonymous: false
Enable Certificate Enrollment: true
CA Type: TAKServer
Signing Keystore File: certs/files/intermediate-ca-signing.jks
Validity Days: 3650
[Edit Security](#)

Authentication Configuration (LDAP)

URL: ldap://192.168.1.115
User String: uid={username},ou=TAK,dc=user,dc=com
Update Interval: 60
Group Prefix:
Service Account DN: cn=admin,dc=user,dc=com
Group Base RDN: ou=TAK
Your Authentication Configuration looks good!

[Edit Authentication](#)
[Test Service Account](#)

TAK Server - LDAP Integration - Connecting our TAK Server

- You can validate if you did it correctly by opening up a browser and entering in: <IP address>:8446
 - The IP address is that of your TAK Server and the specific port will bring you to WebTAK login.
- From there you can enter in the credentials you entered for the users you created on the LDAP service to login.
- If you successfully are logged in you have done everything correctly.

TAK Server - LDAP Integration

- IN and OUT Groups with LDAP:
 - To apply the in/out filters to a group with LDAP you need to create an additional group with the same naming but with either “_READ” or “_WRITE” at the end of the name.

- _READ: out group
- _WRITE: in group

Name	Type	Description
 tak_GroupA	Security Group - Global	
 tak_GroupA_READ	Security Group - Global	OUT Group
 tak_GroupA_WRITE	Security Group - Global	IN Group
 tak_GroupB	Security Group - Global	
 tak_GroupB_READ	Security Group - Global	OUT Group
 tak_GroupB_WRITE	Security Group - Global	IN Group

- If you somehow forget to add a user to a group, they will default to the __ANON__ group when they authenticate to the TAK Server.

Resources:

- <https://www.golinuxcloud.com/install-and-configure-openldap-on-ubuntu-step-by-step/>
- <https://docs.oracle.com/en/operating-systems/oracle-linux/6/admin/ldap-addusers-group.html>
- <https://serverfault.com/questions/514870/how-do-i-authenticate-with-ldap-via-the-command-line>